

Project Report

Fire & Smoke Detection using

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Group No: 07

Course Name: CSE468

Section: 01

1. Project Proposal

Build or Train a deep learning model that can reliably detect fire and smoke at long distances, using optic camera.

1st Target: Gather dataset related to indoor, outdoor, small and large fire and smoke, upto 20k images.

Project Introduction

Uncontrolled fire is hazardous, and can rack up huge costs in damages and loss of life. Traditional detection methods depends on physical sensors, which often has limited coverage and is quite expensive. What our project aims to do is to detect fire and smoke from cameras, in real time, which is cheap to deploy.

To elaborate,. the objective of this project is to design an automated system capable of detecting fire and smoke using image processing techniques. By training a model on a well-prepared dataset containing images of fire and smoke, the system will be able to accurately identify their presence in different scenarios. This type of solution has practical applications in areas such as buildings, early forest fire detection, and industrial safety systems.

2. Week 1 — Project Proposal & Selection

Key Activities & Individual Contributions:

- Several project ideas were discussed and analyzed based on feasibility and scope.
- The fire and smoke detection project was chosen with agreement from all members.
- Responsibilities were distributed among team members.
- I contributed by researching existing studies and related work on fire and smoke detection.
- I examined different image classification and object detection techniques to identify suitable methods.
- I prepared a short document describing the system design and expected results, which was shared with the group.

3. Week 2 — Research & Project Justification

Key Activities & Individual Contributions:

- A shared Git repository was created for collaboration and version control.
- Repository Link: <https://github.com/Starner22/Cse468-Fire-and-Smoke-Detection>
- The team started collecting image datasets for training the model.
- I carried out detailed research on existing detection systems, including CNN and YOLO-based approaches.
- I evaluated available datasets based on quality, size, and relevance.
- I summarized my findings and shared recommendations with the team to guide technical decisions.